1. For your upcoming family reunion, you are in charge of one of the day's activities. You decide to have the family head out in two-person canoes. However, you know that you need to be careful in pairing your family members up, because they can get jealous if they get stuck spending time with one person when they would prefer to spend time with another. Because you are taking CSCI 332, you wonder if you could re-work the stable matching definition from the hospital-student problem to work for your canoe pair problem. You consider asking each family member to order the rest of the family members according to their preference for a canoe partner.

   For example, for $n = 4$ family reunion attendees, you might get the following list:

   Alice: Ben, Cleo, Desiree
   Ben: Cleo, Alice, Desiree
   Cleo: Alice, Ben, Desiree
   Desiree: Alice, Ben, Cleo

   (a) Re-work the definition of stable matching from the hospital-medical student version of the problem to the canoe pair version of the problem. You should describe the input of the problem and the desired output of the problem.

   Solution: The canoe pair stable matching problem is defined as follows:
   Input: A set of n family members, each of whom ranks the other $n-1$ players in order of preference to be their canoe partner. Assume that n is even.
   Output: A matching of the n family members into $n/2$ pairs that is *perfect* (every family member is in exactly one pair) and *stable*: that is, no two family members who are not currently paired up prefer one another over their current pair.
   ■

   (b) Does a stable matching always exist for the canoe pair problem?

   Solution: No.
   ■

   (c) How do you know?

   Solution: To show that there is not always a stable matching, we can give an example input where no such matching exists. Conveninently, the example above is such an input! To see why there is no stable matching, consider every perfect matching possible for this input.
   - $M = \{\{A, B\}, \{C, D\}\}$: in this case, B and C form an unstable pair, since B prefers C over A and C prefers B over D.
   - $M = \{\{A, C\}, \{B, D\}\}$: in this case, A and B form an unstable pair, since A prefers B over C and B prefers A over D.
   - $M = \{\{A, D\}, \{B, C\}\}$: in this case, A and C form an unstable pair, since A prefers C over D and C prefers A over B.
   ■

2. Back to the hospital-student stable matching problem: describe and analyze an efficient algorithm to determine whether a given set of preferences has a *unique* stable matching.

Solution: As we saw in class, the Gale-Shapley algorithm produces the "best" matching for the proposers and the "worst" matching for the proposees. So we can run Gale-Shapley twice, once with hosptials proposing and once with students proposing. If the resulting matchings are the same, then it is the unique stable matching for the input. Otherwise, there are at least two distinct matchings.

Each run of Gale-Shapley takes $O(n^2)$ time, and checking whether the outputs are equal is also $O(n^2)$ time, so the runtime is $O(n^2)$ overall. ■